



Figure 1: Real-time data analytics by InternetLiveStats.com



Figure 2: Statista as a prominent and key portal for statistical data

On WhatsApp, there are 700 million active users. There are more than 1 million new user registrations every month. Around 30 billion messages are sent and 34 billion received every day on WhatsApp. On Facebook, five new profiles are created every second. There are also around 83 billion fake profiles. Around 300 billion photos are uploaded every day by 890 billion daily active users. About 320TB of data is processed daily, with 21 minutes being spent by every user, on an average.

Now, the question is: how to do research on these datasets? Also, which technologies can be used to fetch the real-time datasets? The live streaming data can be fetched using Python, PHP, Perl, Java and many others used for network programming.

Fetching live streaming data from Twitter using Python code

Specific packages named Tweepy and Twitter with Python are required to fetch live tweets from Twitter. After these packages are installed, the Python code will be able to fetch live data from Twitter.

These can be installed using the Pip command as follows:

```
$ python -m pip install tweepy
$ python -m pip install Twitter
```

The code to fetch live tweets from Twitter is:

```
from tweepy import Stream
from tweepy import OAuthHandler
from tweepy.streaming import StreamListener
my_app_consumerkey =
'XXXXXXXXXXXXXXXXXXXXXXXXX'
my_app_consumersecret = '
XXXXXXXXXXXXXXXXXXXXXXXXX '
my_app_access_token = '
XXXXXXXXXXXXXXXXXXXXXXXXX '
my_app_accesssecret = '
XXXXXXXXXXXXXXXXXXXXXXXXX '
class TweetListener(StreamListener):
    def on_data(self, mydata):
        print mydata
        return True
    def on_error(self, status):
        print status
auth = OAuthHandler(my_app_consumerkey,
my_app_consumersecret)
auth.set_my_app_access_token(my_app_
access_token, my_app_accesssecret)
stream = Stream(auth, TweetListener())
stream.filter(track=[Name of the Celebrity
or Movie or Person'])
```

After execution of this script, the output dataset is fetched in JSON file format. The JSON file can be parsed using the OpenRefine tool in the XML, CSV or any other readable format by the data mining and machine learning tools.

OpenRefine is a powerful and effective tool used for processing the Big Data and JSON file formats.

In a similar way, the timeline of any person or Twitter ID can be fetched using the following code:

```
import tweepy
import time
my_app_consumerkey = 'XXXXXXXXXXXXXXXXX'
my_app_consumersecret = 'XXXXXXXXXXXXXXXXX '
my_app_access_token = 'XXXXXXXXXXXXXXXXX '
my_app_accesssecret = 'XXXXXXXXXXXXXXXXX '
auth = tweepy.auth.OAuthHandler(my_app_consumerkey, my_app_
consumersecret)
auth.set_my_app_access_token(my_app_access_token, my_app_
accesssecret)
api = tweepy.API(auth)
list= open('Twitter.txt','w')
if(api.verify_credentials):
    print 'Connected to Twitter Server'
    currentuser = tweepy.Cursor(api.followers, screen_
name='gauravkumarin').item()
    while True:
        try:
            u = next(currentuser)
```